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**Please fill in answers in the space supplied and show all your work to receive full credit.
Useful Tables at end of Exam.**

1. a) (10 pts) State whether the following statements are **True of False**
- An ideal gas at a given state expands to a fixed final volume at constant pressure or at constant temperature. Work is smaller in constant pressure case.
 - A two-phase liquid-vapor mixture with equal volumes of saturated vapor and saturated liquid has a quality of less than 0.5.
 - As temperature decreases from the critical temperature, the values of the specific volume of saturated vapor and saturated liquid moves apart from each other.
 - Internal energy is an intensive property.
 - Specific volume of compressed liquid can be approximated to specific volume of saturated liquid at the same pressure.
- b) (5 pts) The gage pressure of an automobile tire is measured to be 30psig before a trip and 32psig after the trip at a location where the atmospheric pressure is 14psia. Assuming the volume of the tire remains constant and the air temperature before the trip is 25°C, determine the air temperature in the tire after the trip. (Air is ideal gas)
- c) (5pts) What would be the specific volume of saturated solid carbon dioxide at the same temperature, greater or less than saturated vapor value? Why?

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2. (50 pts) 5kg of water at contained in a piston–cylinder assembly undergoes two processes in series from an initial state where the temperature is 120°C and 10 bar.

- **Process 1–2** System is heated at constant pressure to a saturated vapor state.
- **Process 2–3** System is cooled at constant volume until it reaches to 120°C.

Kinetic and potential effects can be ignored.

- a) (3 pts) Draw a schematic of the system and label the system boundary.
- b) (8 pts) Carefully sketch the process on a:
 - (1) P-v diagram (Pressure-specific volume)
 - (2) T-v diagram (Temperature-specific volume)
- c) (5 pts) State the engineering model
- d) (2 pts) Write the simplified energy balance
- e) (15 pts) Calculate the work transferred in kJ. Is work done on the system or by the system?
- f) (15 pts) Calculate the heat transferred in kJ. Is heat transferred to the system or from the system?
- g) (2 pts) Calculate the mass of vapor in each state in kg.

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3. (30 points) A piston cylinder device initially contains 0.5m³ of air at 80kPa and 200°C. The air is now compressed to 0.1m³ in such a way that the temperature inside the cylinder remains constant. The air is modeled as an ideal gas and kinetic and potential energy effects are negligible.
- (2 pts) Draw a schematic of the system and label the system boundary.
 - (2 pts) State the engineering model
 - (3 pts) Write the simplified energy balance and calculate compressibility factor (Z) for ideal gas and not ideal gas state 1.
 - (3pts) Draw P-v diagram (Pressure-specific volume)
 - (10 pts) Determine the work (kJ)
 - (10 pts) Determine the heat (kJ)

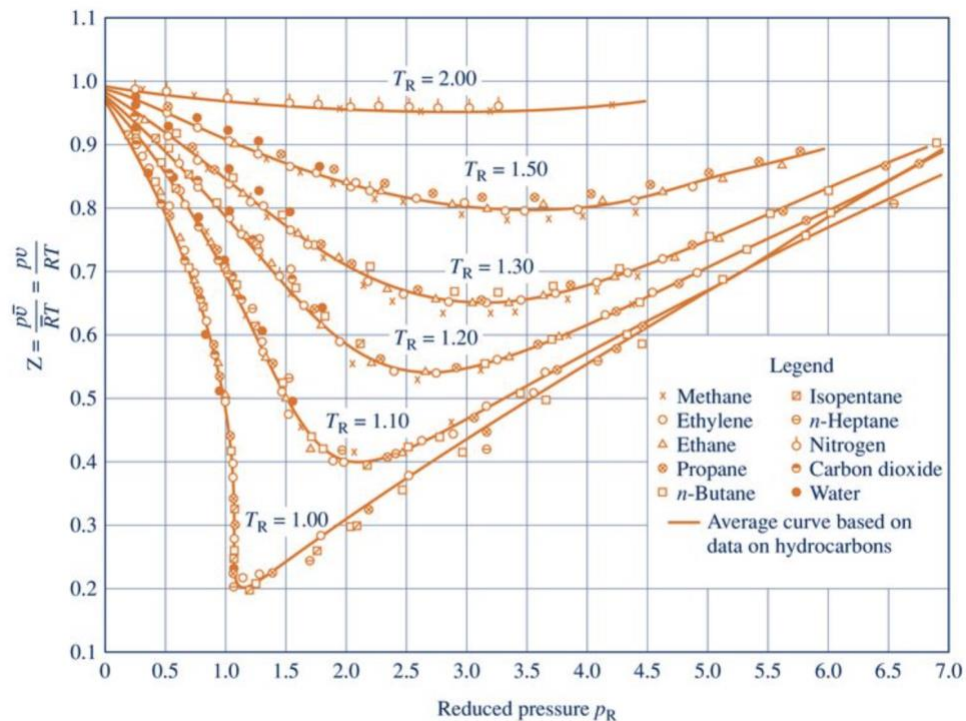


FIG. 3.12 Generalized compressibility chart for various gases.

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